

Drone Contingency Management Model Overview:

Modelling drone contingency actions in a shared airspace

Why - Understanding drone safety in shared airspace

- **Consider:** Why can't drones and piloted aircraft fight fires at the same time?
- **Answer:** We don't yet know how to manage conflicts and drone-related hazards in the shared airspace with access restrictions
- The goal of this model is to **better understand how to mitigate these hazards**

PC: FBI/KTLA, ktla.com/news/california/wildfires/fbi-looking-for-pilot-of-drone-that-grounded-plane-battling-palisades-fire/



What are we trying to do?

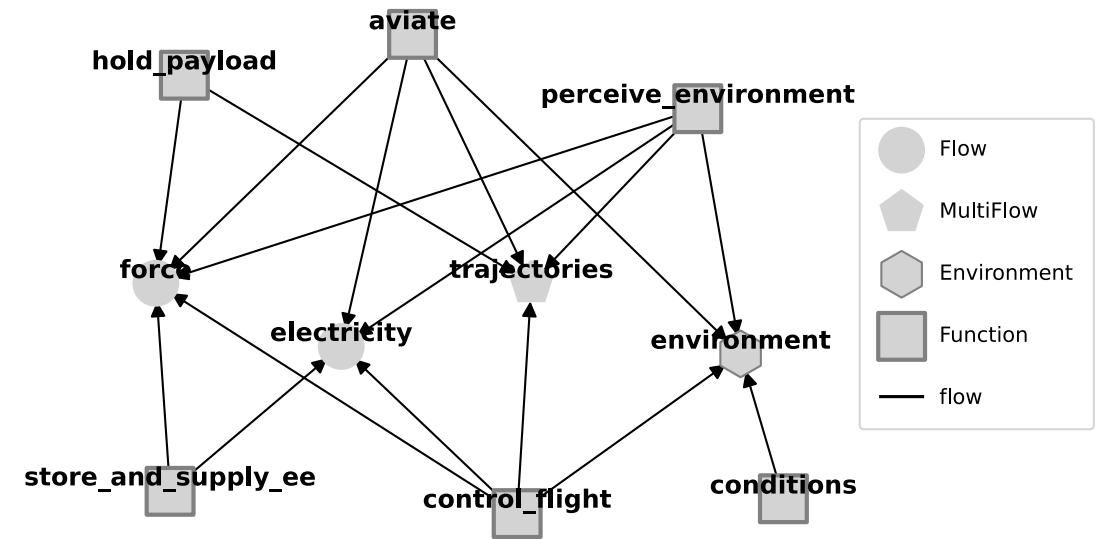
Resilience models help us understand how well a system will mitigate hazardous scenarios. In this case, we want to:

1. Evaluate the ability of proximity to threat functionality to enable safe operations in drones
2. Identify backup/redundant battery storage requirements to mitigate battery depletion faults
3. Better understand how a given mission or Concept of Operations can affect resilience(s)

To do this, we adapt the base `airspace1ib` library build with `fmdtools` to represent drone behavior, and its interactions with its environment in the relevant scenarios.

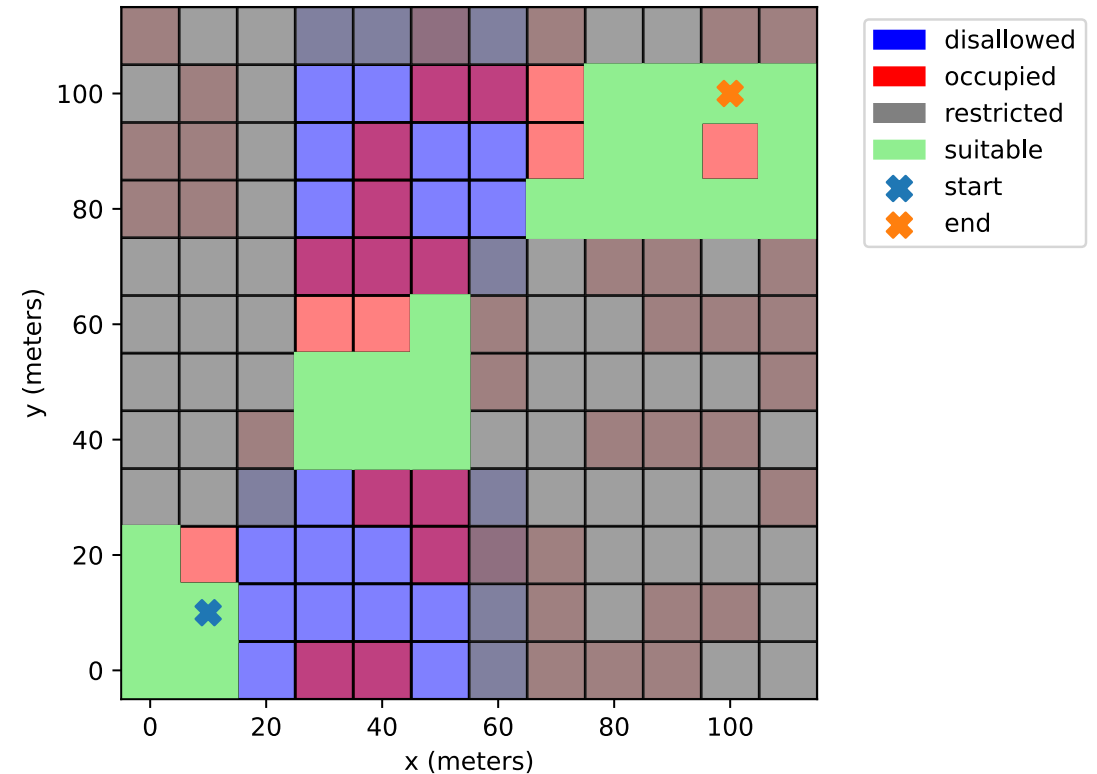
Setup: Model Structure

- **aviate**: movement of the drone through the air
 - Alters true drone trajectory
- **control_flight**: path planning and control
 - Alters desired drone trajectory
- **store_and_supply_ee**: battery/energy storage
 - Provides electricity to other functions
- **perceive_environment**: Perception of drone location/environment
 - Determines perceived trajectory from true
- **hold_payload**: Force balance/structure
- **conditions**: External update of environment
 - Determines external drone location(s)



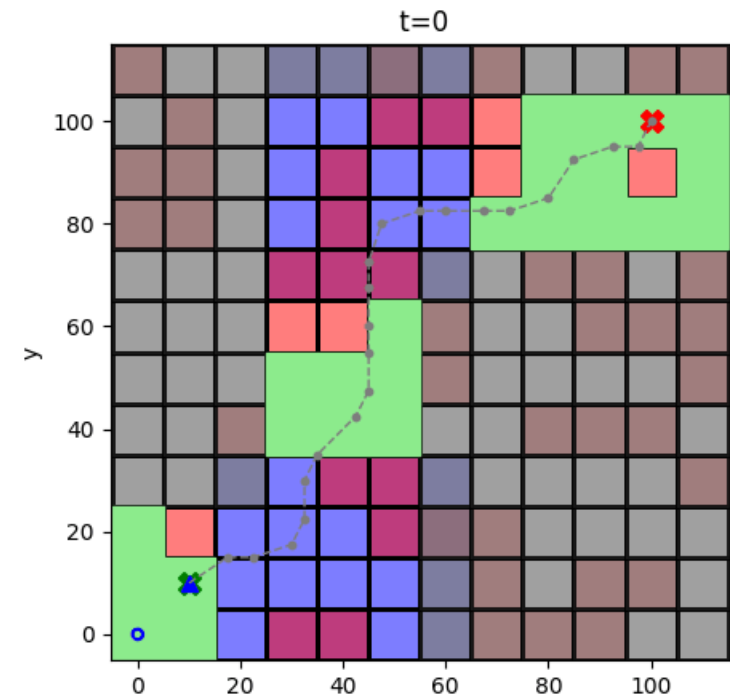
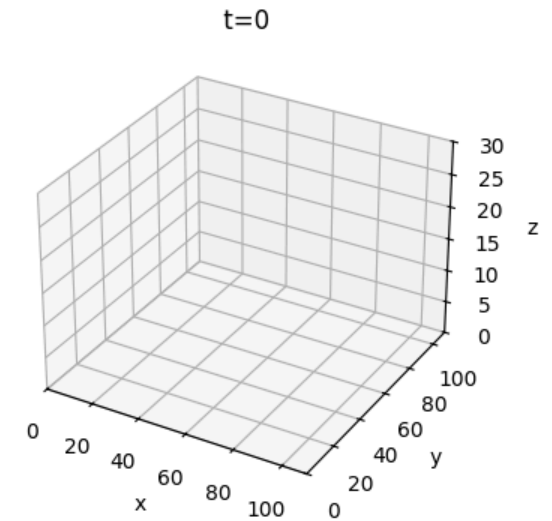
Setup: Environment and Mission

- Drone's mission is to fly from lower-left start point to upper-right end point
- Flies through area not designated "Restricted" (in gray)
- Can land in green "suitable" areas in emergencies
- Cannot land in Occupied (red) and Disallowed (blue) areas



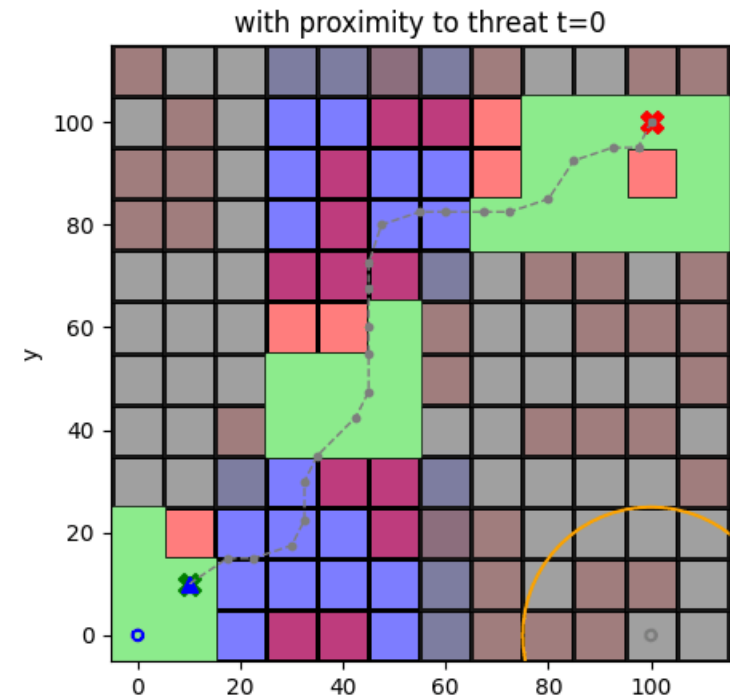
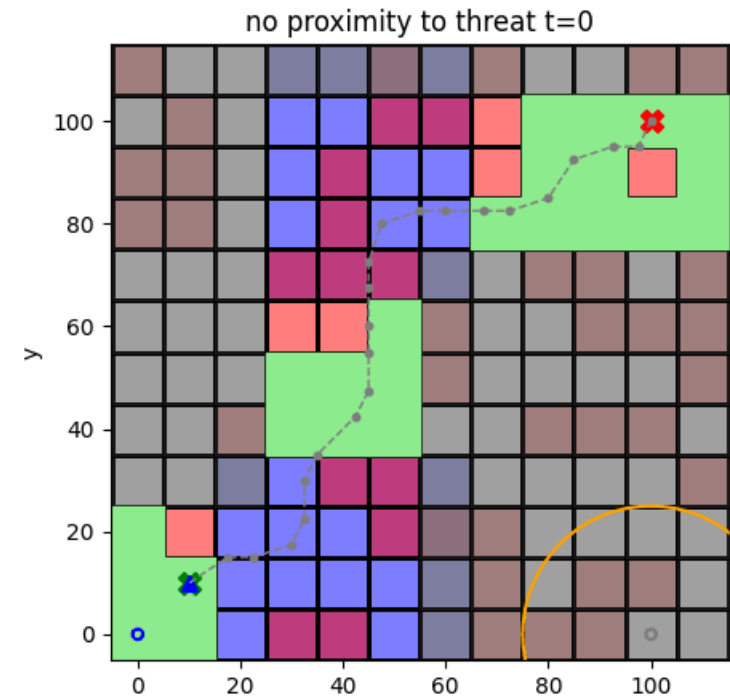
Setup: Path planning and reconfiguration

- Drone plans path to goal while avoiding restricted airspace and minimizing landing risks
- Drone at a constant height
- Re-plans when hazardous conditions are identified:
 - **airspace intrusion or low battery**



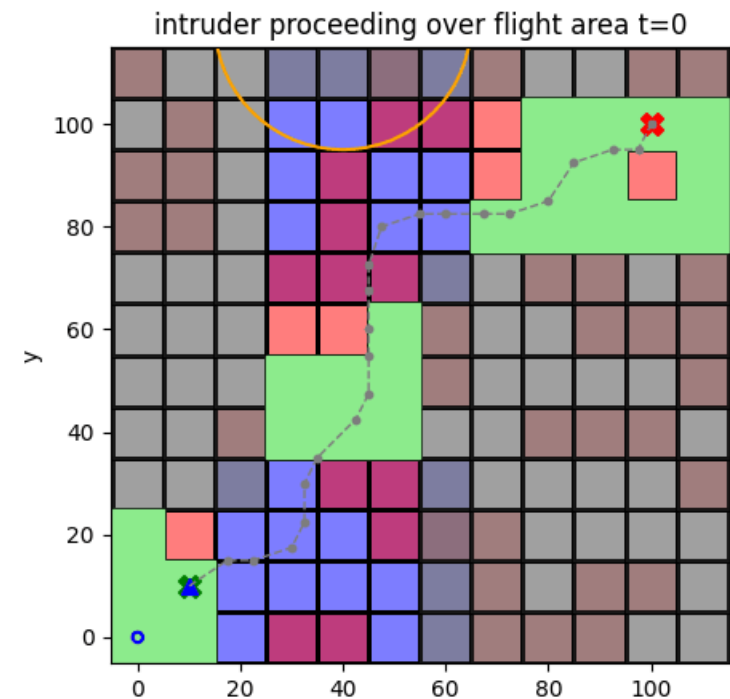
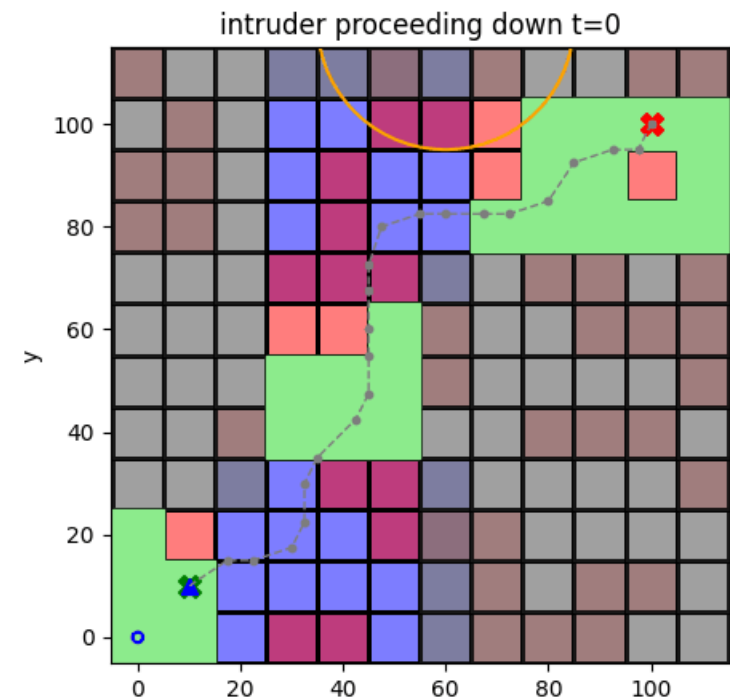
How resilient is the drone to airspace intrusion?

- Without proximity to threat functionality, drone may fly into errant intruding drone
- Proximity to threat functionality causes a pause in mission as well as mission re-planning



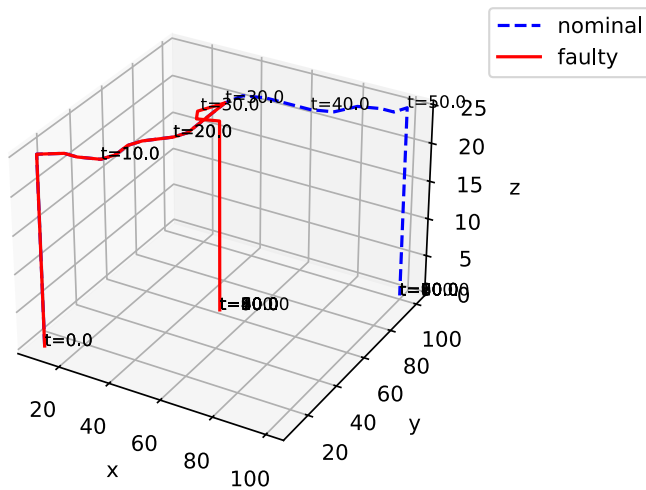
Would the drone still be resilient to airspace intrusion in a different scenario?

- If the intruder is easily avoidable, we can plan around it
- If not easily avoidable, this can cause issues
 - No logic for "running away"
 - May replan into restricted flight area

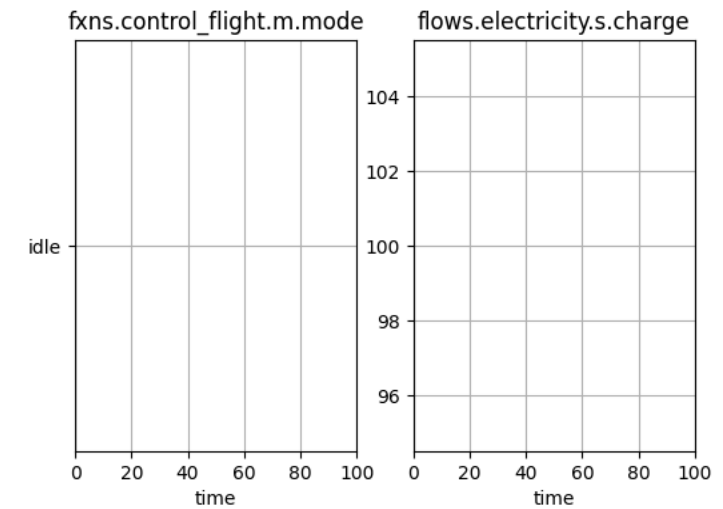


How resilient is the drone to battery depletion?

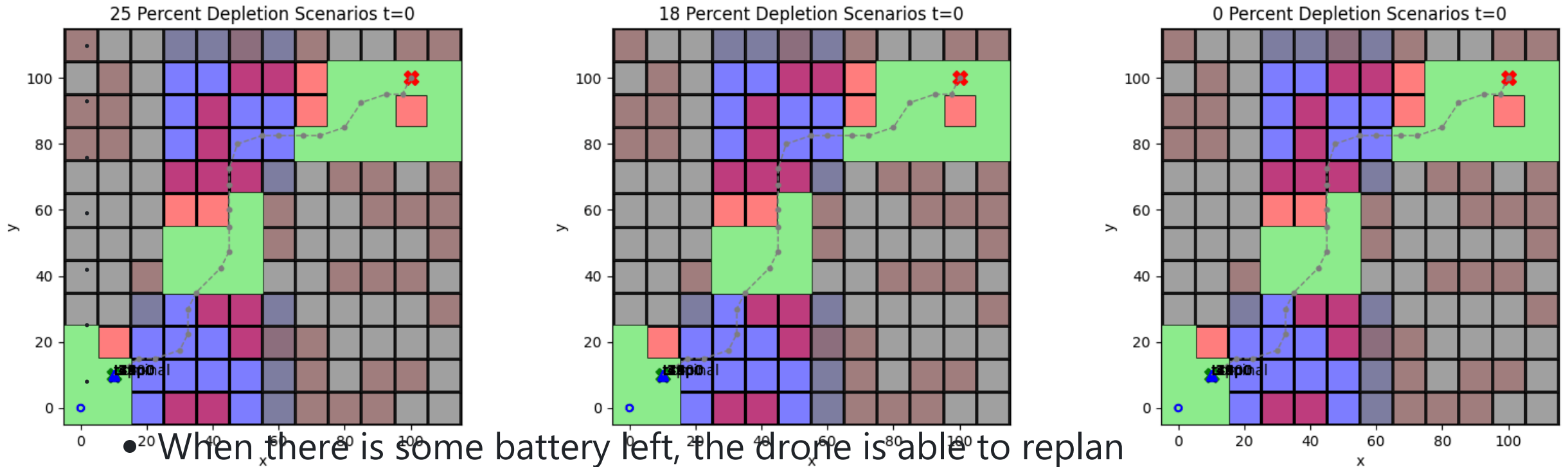
Trajectory, Depletion Scenario



- When SoC is below a threshold, flies to closest suitable location



How resilient is the drone to battery depletion overall?



- When there is some battery left, the drone is able to replan
- Otherwise, the drone may not make it to a suitable landing spot or may crash

How resilient is the drone to battery depletion overall?

Scenario SoC	mission complete	unsuitable site	disallowed site	occupied site	damaged	crashed
0%	10	90	40	10	100	100
18%	10	40	20	20	0	0
25%	10	0	0	0	0	0

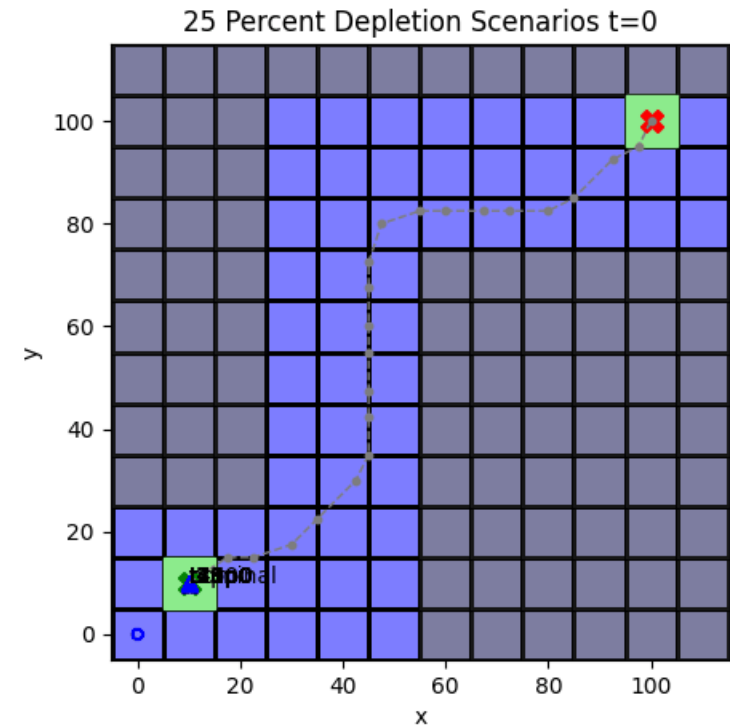
- Adequate mission recovery at 25% SoC - all suitable landings
- More unsuitable landings at 18% SoC (at 15% the drone lands immediately)
- At 0% the drone crashes

Would the drone still be resilient in a different mission?

No!, consider the 25% depletion scenarios at right:

- Flying over disallowed area
- Statistics: 20% mission complete, 40% in disallowed locations

Effectiveness of drone safety features is mission and scenario-dependent



Analysis Conclusions

Important Drone Features:

- Proximity to Threat functionality can improve drone resilience to flightplan intrusions
- Battery monitoring can also help improve drone resilience to battery faults
- Both of these require **in-flight environmental risk perception** and **risk-aware replanning** to be used effectively

But, Execution Matters...

- Need to have adequate battery redundancy to respond effectively--which may be different depending on the type of mission (are there bail-out points?)
- Re-planning to avoid approaching drones is more difficult than static ones

Potential Future Work

- A variety of bug fixes and feature improvements
 - More robust logic for avoiding errant drones
 - More robust logic for battery depletion path planning
 - ...
- Can we use this to determine whether a drone is "safe enough" for a mission of interest
 - Mission: Start, end location and map
 - Is the battery adequate?
 - Can we bail out if another vehicle flies through?
- Evaluating other interesting hazardous scenarios
 - Wind, etc.

Drone Resilience Library Conclusions

1. Resilience models help us understand **dynamic** aspects of safety, such as...
 - **Dynamical system behaviors** (power draw, flight behavior, etc.) that key system functionality, performance, and safety is ultimately based on
 - **When/where a hazard occurs** in a dynamic mission/environmental context
 - **Dynamic resources (e.g., energy storage)** the system can leverage to mitigate those faults and their effectiveness
2. This Drone Library and resulting Contingency Management model was implemented in **fmdtools**, which provides straightforward interfaces for building, simulating, and analyzing resilience models in **Python**
 - Most analyses and visuals used fmdtools constructs (modeling classes, simulation and analysis methods) directly rather than writing custom code
 - Underlying methodology and code constructs **can be adapted to a range of applications** (autonomous vehicles, space, etc), not just drones